

A Sparse Non-Unimodular Equality Case for the Complex Spencer Problem in Dimension Four

Autonomous discovery packet for arXiv:2602.12868

13 August 2026

Verdict. Candidate full solution, likely valid, to the source’s explicit equality-case question. This note constructs non-unimodular rows attaining the sharp value $\sqrt{4} = 2$. It does *not* prove the universal $n = 4$ instance of Conjecture 1.2.

1 Source question

Tomasz Kobos and Marin Varivoda, *A Complex Analogue of Spencer’s Six Standard Deviations Theorem and the Complex Banach–Mazur Distance*, arXiv:2602.12868 (2026), formulate Conjecture 1.2: if $a_1, \dots, a_n \in \mathbb{C}^n$ satisfy $\|a_i\|_\infty \leq 1$, then some $x \in \mathbb{T}^n$ satisfies $|\langle x, a_i \rangle| \leq \sqrt{n}$ for all i . They prove it for $n = 2, 3$. At the end of page 26 they write:

“We do not know whether, as in the case $n = 3$, there also exist non-unimodular vectors achieving equality.”

Here “unimodular” means that every coordinate has modulus one, and the source uses $\langle x, y \rangle = \sum_j x_j \overline{y_j}$. The complete source context is reproduced below.

We believe that establishing Conjecture (1.2) for $n \geq 4$ will be highly non-trivial, even in its weaker version, where the vectors $a_1, \dots, a_n \in \mathbb{C}^n$ are assumed to be unimodular. Under this assumption, the corresponding higher-dimensional toric bodies centered in vectors a_i also differ only by translation. However, for example in dimension $n = 4$, there exists a one-parameter infinite family of pairwise non-equivalent complex Hadamard matrices, each achieving equality in (1.2). Therefore, since the bound $\sqrt{n}$ in the conjecture is sharp, any proof for $n = 4$ must necessarily account for this infinite family. We do not know whether, as in the case $n = 3$, there also exist non-unimodular vectors achieving equality. More broadly, we suspect that Conjecture 1.2 is deeply intertwined with the geometry of $\mathbb{C}^n$, and in particular, with the classification of $n \times n$ complex Hadamard matrices.

2 Result

For a row system $A = (a_i)_{i=1}^4$, put

$$D(A) := \min_{x \in \mathbb{T}^4} \max_{1 \leq i \leq 4} |\langle x, a_i \rangle|.$$

The source calls A an equality case in dimension four when $D(A) = 2$.

Proof intuition

Take the four sign patterns of two variables and append a zero coordinate. The resulting correlations are the four vertices $1 \pm z \pm w$ of a planar parallelogram translated by 1. If all four vertices entered the open disk of radius two, elementary expansion would force the three directions $1, z, w$ to have pairwise line angles strictly larger than 60° . Three lines cannot be packed that way into a projective circle of circumference 180° . Cube-root directions attain the boundary exactly.

Theorem 1 (Non-unimodular $n = 4$ extremizer). *For $(\varepsilon, \delta) \in \{\pm 1\}^2$, let*

$$a_{\varepsilon, \delta} = (1, \varepsilon, \delta, 0) \in \mathbb{C}^4.$$

Then

$$\min_{x \in \mathbb{T}^4} \max_{\varepsilon, \delta \in \{\pm 1\}} |\langle x, a_{\varepsilon, \delta} \rangle| = 2.$$

Consequently, dimension four admits a non-unimodular equality case for Conjecture 1.2.

Proof. Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{T}^4$. Multiplying all correlations by $\overline{x_1}$ does not change their moduli, so set $z = x_2 \overline{x_1}$ and $w = x_3 \overline{x_1}$. Thus $|z| = |w| = 1$ and

$$|\langle x, a_{\varepsilon, \delta} \rangle| = |1 + \varepsilon z + \delta w|.$$

Write

$$a = \operatorname{Re} z, \quad b = \operatorname{Re} w, \quad c = \operatorname{Re}(z\overline{w}).$$

Expansion gives

$$|1 + \varepsilon z + \delta w|^2 = 3 + 2q_{\varepsilon, \delta}, \quad q_{\varepsilon, \delta} = \varepsilon a + \delta b + \varepsilon \delta c. \quad (1)$$

Suppose, for contradiction, that all four moduli were strictly less than 2. Then every $q_{\varepsilon, \delta} < 1/2$. Adding the two inequalities with a fixed first sign gives

$$q_{+,+} + q_{+,-} = 2a < 1, \quad q_{-,+} + q_{-,-} = -2a < 1,$$

and hence $|a| < 1/2$. Fixing the second sign in the same way yields $|b| < 1/2$. Finally,

$$q_{+,+} + q_{-,-} = 2c < 1, \quad q_{+,-} + q_{-,+} = -2c < 1,$$

so $|c| < 1/2$.

Regard $1, z, w$ as unit vectors in the real plane. Their three pairwise real inner products are a, b, c . Passing to unoriented lines places their arguments on the projective circle $\mathbb{R}/\pi\mathbb{Z}$. Three points divide this circle of circumference π into three gaps, one of which is at most $\pi/3$. The corresponding two lines therefore have acute angle at most $\pi/3$, so the absolute value of their inner product is at least $\cos(\pi/3) = 1/2$. This contradicts $|a|, |b|, |c| < 1/2$. Hence every $x \in \mathbb{T}^4$ has at least one correlation of modulus at least 2.

For the reverse inequality let $\omega = e^{2\pi i/3}$ and take $x = (1, \omega, \omega^2, 1)$. Since $1 + \omega + \omega^2 = 0$, the four correlation moduli are

$$|1 + \omega + \omega^2|, \quad |1 + \omega - \omega^2|, \quad |1 - \omega + \omega^2|, \quad |1 - \omega - \omega^2| = 0, 2, 2, 2.$$

Thus $D(A) \leq 2$, completing the proof. Each row has a zero fourth coordinate, so none is unimodular. $\square$

3 Verification

The proof is exact and computation-free. The accompanying script `code/verify_extremizer.py` independently checks:

- the four exact cube-root witness values 0, 2, 2, 2;
- one million uniformly sampled phase pairs, all satisfying the lower bound to floating-point tolerance; and
- a differential-evolution minimization returning 2 to better than 10^{-10} .

The key identity (1) can also be checked directly term by term. No numerical statement is used in the proof.

4 Scope, novelty, and review recommendation

This theorem fully answers the narrow existence question quoted above. It does not classify equality cases, prove Conjecture 1.2 for arbitrary four-row systems, or settle either Banach–Mazur conjecture in the source.

On 13 August 2026, bounded searches used arXiv:2602.12868, the exact paper title, the exact phrase “non-unimodular vectors achieving equality,” and combinations of “complex Spencer,” “dimension 4,” and “non-unimodular equality.” They found the source preprint and mirrors but no later resolution, follow-up, or erratum. This supports, but does not certify, novelty.

Human review should focus on (i) the strict-inequality step from four open radius-two disks to $|a|, |b|, |c| < 1/2$, (ii) agreement with the source inner product convention, and (iii) whether the sparse Walsh configuration is known under another formulation. Subject to those checks, this is ready to report to the authors as a short affirmative answer to their equality-case question.

References

- [1] T. Kobos and M. Varivoda, *A Complex Analogue of Spencer’s Six Standard Deviations Theorem and the Complex Banach–Mazur Distance*, arXiv:2602.12868, 2026.